

[LAB Report - 2]

Members: Imtiaz Hossain (23101137)

Siam Rahman Saajin (23101312)

Abdus Sami Rasi (23201521)

Md. Shahriar Kabir (24141263)

Course : CSE260

Section : 01

Group : 01

Experiment Name:

Experiment # 4: Design and Implementation of 4 bit Parallel Binary Adder

Experiment # 5: Implementation of 4-bit magnitude comparator

Experiment # 6: Design circuits using encoder & decoder.

Experiment # 7: Function Implementation using MUX

Required components for Lab-4:

1. IC 7408
2. IC 7402
3. IC 7486
4. IC 7483

Required components for Lab-5:

1. IC 7408
2. IC 7432
3. IC 7404
4. IC 4077

Required components for Lab-6:

1. IC 74138
2. IC 74148

Required components for Lab-7:

1. IC 74153
2. IC 7408
3. IC 7432
4. IC 7404

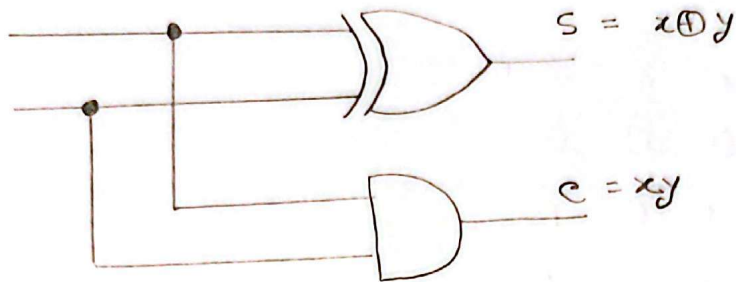
[Experiment-4]

(LAB-4)

24.07.2025

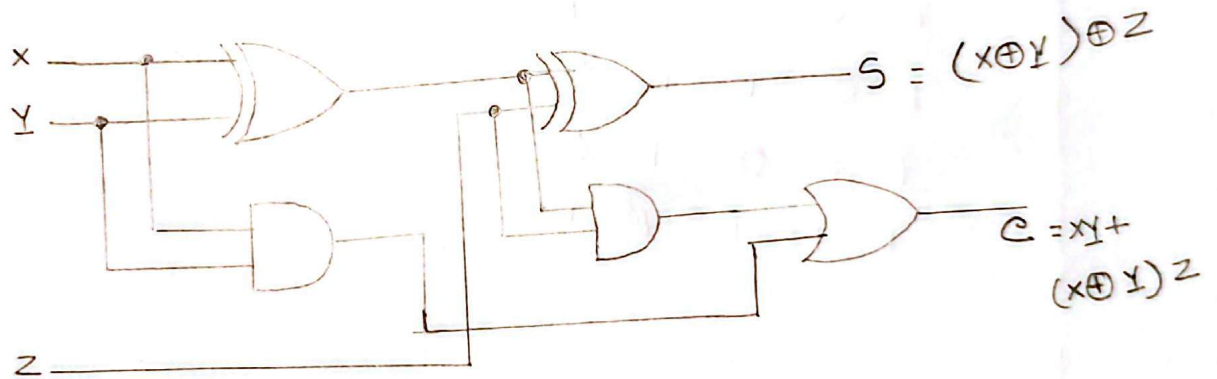
[Experimental setup]

Half Adder circuit



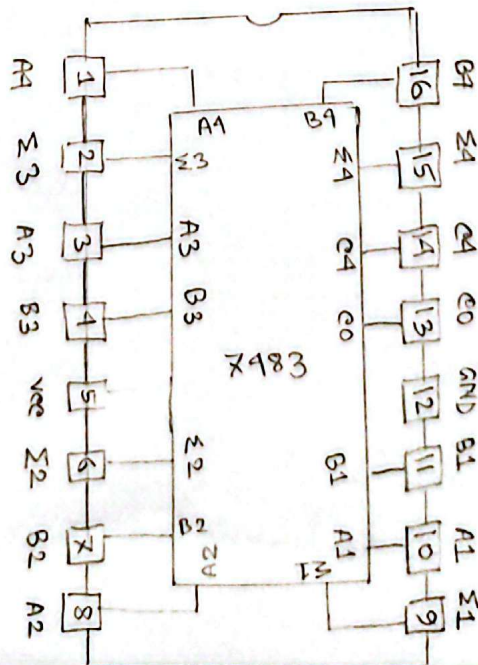
X	Y	C	S
0	0	0	0
0	1	0	1
1	0	0	1
1	1	1	0

Full Adder circuit:



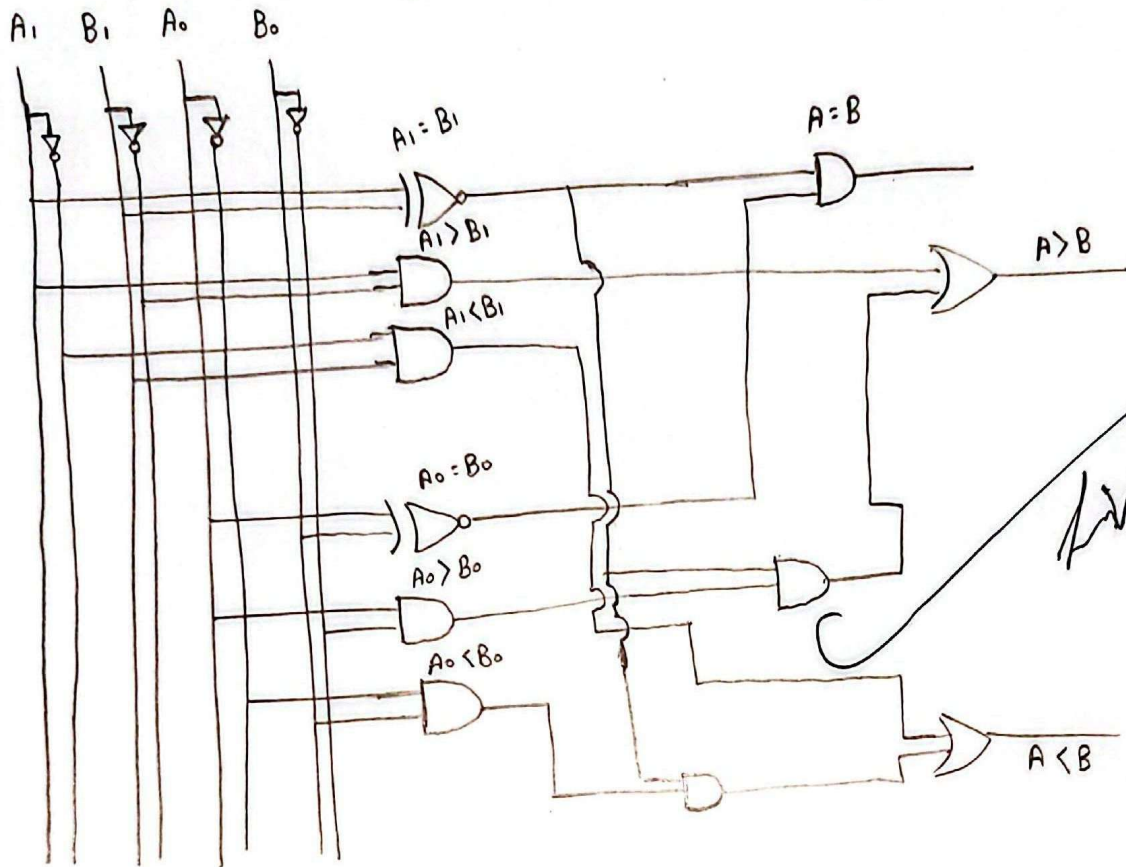
A four bit parallel adder circuit:

(IC - 7483)



24/07/2025

[Experimental Setup] [Experiment - 5]



[Experiment-4]

[Result]

~~A four bit parallel adder subtractor:~~

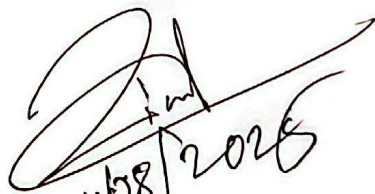
Full adder truth-table

x	y	z	c	s
0	0	0	0	0
0	0	1	0	1
0	1	0	0	1
0	1	1	1	0
1	0	0	0	1
1	0	1	1	0
1	1	0	1	0
1	1	1	1	1

[Experiment-6]

[Result]

Inputs				Expected output				Active Low outputs		
Minterm	C	B	A	Minterm	D _L	D _I	D _O	D _L	D _I	D _O
0	0	0	0	0	0	0	0	1	1	1
1	0	0	1	7	1	1	1	0	0	0
2	0	1	0	6	1	1	0	0	0	1
3	0	1	1	5	1	0	1	0	1	0
4	1	0	0	4	1	0	0	0	1	1
5	1	0	1	3	0	1	1	1	0	0
6	1	1	0	2	0	1	0	1	0	1
7	1	1	1	1	0	0	1	1	1	0


14/08/2025

[Experiment -7]

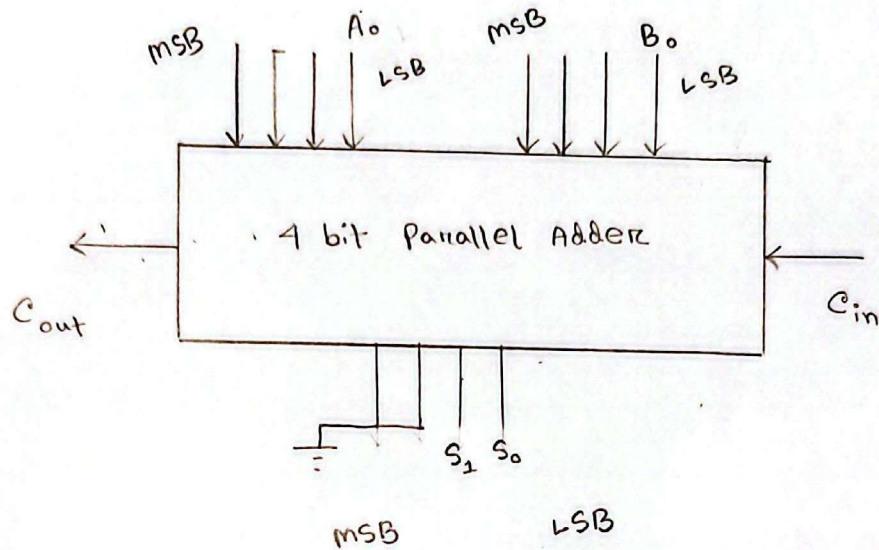
[Result]

	A	B	C(S1)	D(S0)	I3	I2	I1	I0	F
0	0	0	0	0	0	0	0	0	0
1	0	0	0	1	0	0	1	0	1
2	0	0	1	0	0	1	0	0	1
3	0	0	1	1	0	0	0	0	0
4	0	1	0	0	0	0	0	1	1
5	0	1	0	1	0	0	1	0	1
6	0	1	1	0	0	1	0	0	1
7	0	1	1	1	0	0	0	0	0
8	1	0	0	0	0	0	0	0	0
9	1	0	0	1	0	0	0	1	0
10	1	0	1	0	0	0	0	0	0
11	1	0	1	1	0	0	0	0	0
12	1	1	0	0	0	0	0	0	1
13	1	1	0	1	0	0	0	1	0
14	1	1	1	0	0	0	0	0	0
15	1	1	1	1	1	0	0	0	0

[Signature]
24.8.25

Answer to the question no 1

[Discussion]



Answer to the question no 2

